

Parabolas

Montague Hamilton

2 October 2026

1 Questions

1. **Describe fully and test:** $y = -3(x + 2)^2 + 5$ - **vertex, opening direction, shape and the y-intercept.**

The vertex will occur at $(-2, 5)$, as

$$y = -3((-2) + 2)^2 + 5$$

$$y = -3(0)^2 + 5$$

$$y = 5$$

The opening direction will be downwards, as the y values get inverted by the negative in front of the expression, $-3(x + 2)^2$. The shape will be quite narrow, as for every addition to x, y changes by 3 (shrinks in this case). The y-intercept occurs when $x = 0$, so

$$y = -3((0) + 2)^2 + 5$$

$$y = -3 \cdot 4 + 5$$

$$y = -7$$

The y-intercept occurs at $(0, -7)$.

2. **Describe fully:** $y = 2 \sin(3x) - 4$ - **amplitude, midline and period.**

The amplitude will be 2, as for every value of $\sin(3x)$, y is twice as big as otherwise. The midline will be -4 , as the last number added or subtracted from the rest of the function shifts every y output by that amount. The period will be 6π , as the function will cycle 3 times faster than otherwise.